

Zero-Padded Root- ℓ^1 Transport and Its Sharp Bridge Clock

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Abstract

The noisy modulus head and the finite counterloop shell may have unequal cardinalities, so an equal-rank matching criterion is ill typed. We repair it by padding the smaller multiset with zeros, which does not change positive power moments. The optimal padded root- ℓ^1 cost controls every moment by a Lipschitz power bound and hence controls the weighted head constituent. For a logarithmic bridge clock with $BR > 1$, the resulting power-rate threshold is $\gamma > a \log(BR)$. A single radial pair proves that strictness is sharp: equality need not vanish. At the minimal bridge clock and the limiting monodromy radius the threshold is $0.6729348509\dots$. The theorem is a sufficient interface; no actual matching for the noisy modulus head is proved.

1 Padded matching

Let $X = \{x_1, \dots, x_p\}$ and $Y = \{y_1, \dots, y_q\}$ be finite multisets. Pad the smaller one with zeros until both have size $N = \max\{p, q\}$, and put

$$d_1^0(X, Y) = \min_{\pi \in S_N} \sum_{j=1}^N |x_j - y_{\pi(j)}|.$$

Zero padding leaves $\sum x_j^n$ and $\sum y_j^n$ unchanged for every $n \geq 1$.

Theorem 1.1. *If $|x_j|, |y_j| \leq B$, then for every $n \geq 1$,*

$$\left| \sum_j x_j^n - \sum_j y_j^n \right| \leq nB^{n-1} d_1^0(X, Y).$$

Consequently, for $m \geq 3$,

$$D_m(R) := \sum_{n=2}^{m-1} \frac{\left| \sum_j x_j^n - \sum_j y_j^n \right| R^n}{n} \leq d_1^0(X, Y) R \sum_{j=1}^{m-2} (BR)^j.$$

Proof. Choose an optimal padded matching and use

$$|x^n - y^n| \leq nB^{n-1} |x - y|.$$

Sum over matched pairs, then over orders. The padding contributes zero to all moments. $\square$

2 Sharp logarithmic rate

Theorem 2.1. *Assume $BR > 1$, $m_\sigma = \lceil a \log(1/\sigma) \rceil$, and $d_1^0(X_\sigma, Y_\sigma) = O(\sigma^\gamma)$. The transport bound tends to zero if $\gamma > a \log(BR)$. This exponent is sharp for the information class: with $X_\sigma = \{B\}$, $Y_\sigma = \{B - \sigma^\gamma\}$, the budget is*

$$\Theta(\sigma^\gamma (BR)^{m_\sigma}).$$

It is $\Theta(1)$ at equality and diverges below the threshold.

Proof. The geometric sum in Theorem 1.1 is $\Theta((BR)^{m_\sigma})$. For the radial pair, $m_\sigma \sigma^\gamma \rightarrow 0$, so uniformly through the moving window

$$B^n - (B - \sigma^\gamma)^n = nB^{n-1}\sigma^\gamma\{1 + o(1)\}.$$

Substitution gives the asserted asymptotic and all three regimes. □

3 Target thresholds and protocol

Let $a_* = 1/\log(10/7)$ and $\beta = (0.85\sqrt{\lambda})^{-1} = 0.9080523604\dots$. Then

$$a_* \log(\beta R) = 0.6729348509145321\dots$$

If only the global scaled cap $B = 1/0.85$ is used, the threshold is

$$a_* \log(R/0.85) = 1.399008185460602\dots$$

The computation checks the finite matching inequality and radial saturation rows; it does not use archived noisy roots.

Remark 3.1. An actual root matching with the required decay rate is not known. The theorem supplies only the D_σ constituent and cannot replace the full-trace constituent E_σ . Gates A–E remain false/open.